

UNNS Color Confinement - First Diagnostic Report

Static spectrum + flux-tube profile + UNNS mapping

Report status: Formalized first diagnostic. **Boundary condition:** this report does not claim that UNNS derives QCD confinement; it formalizes a grounded structural mapping from the current data pack.

Executive conclusion

Color confinement can be used as a physical benchmark for **admissibility-by-closure**: colored constituents may be internally real, but isolated colored externalization is not the admissible output. The observable route is tensioned, localized, and repaired into color-neutral composite channels.

internal colored constituent -> attempted separation -> localized route tension -> threshold / repair region -> admissible color-neutral composite

1. Data status

Static-source energy levels

Metric	Value
Rows	57
Separation range	0.70686 to 1.6065 fm
Data status	model reconstruction; not raw data
Minimum gap V1 - V0	0.078378 GeV
Minimum gap V2 - V1	0.040918 GeV

Interpretation boundary: the static-source levels are model-reconstructed from reported Hamiltonian parameters. They are useful for a first structural diagnostic but must not be treated as raw measured GEVP point data.

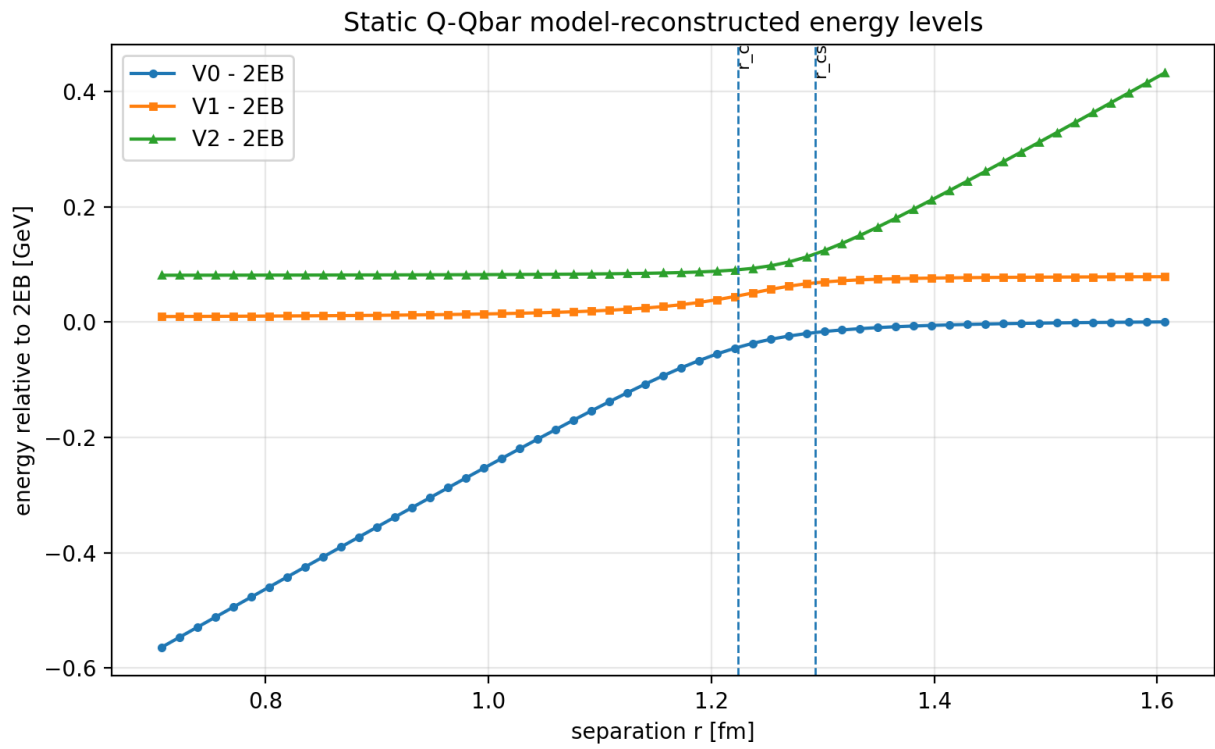
String-breaking thresholds

Threshold	Channel	r [fm]	Nearest r [fm]	gap01 [GeV]	gap12 [GeV]
r_c	light static light threshold	1.224 +/- 0.015	1.22094	0.089700	0.045937
r_cs	strange static strange threshold	1.293 +/- 0.016	1.28520	0.086418	0.046390

Flux-tube pointwise profile

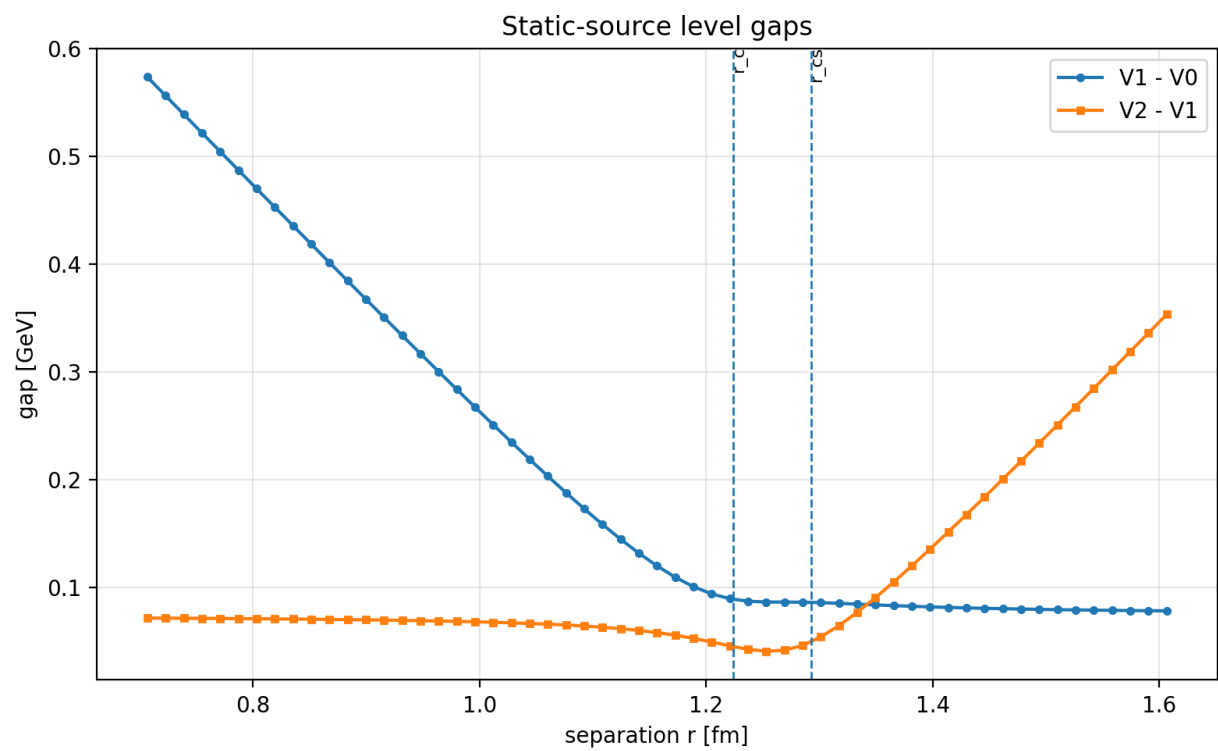
Metric	Value
Rows	31
Transverse range	-1.107463 to 1.107463 fm
Static-source separation	0.738309 fm
Data status	author ancillary pointwise data
Peak Ex FULL	0.343135 GeV ²
Peak Ex nonperturbative	0.244566 GeV ²
Estimated NP FWHM	0.551469 fm

Figure 1. Static Q-Qbar model-reconstructed energy levels



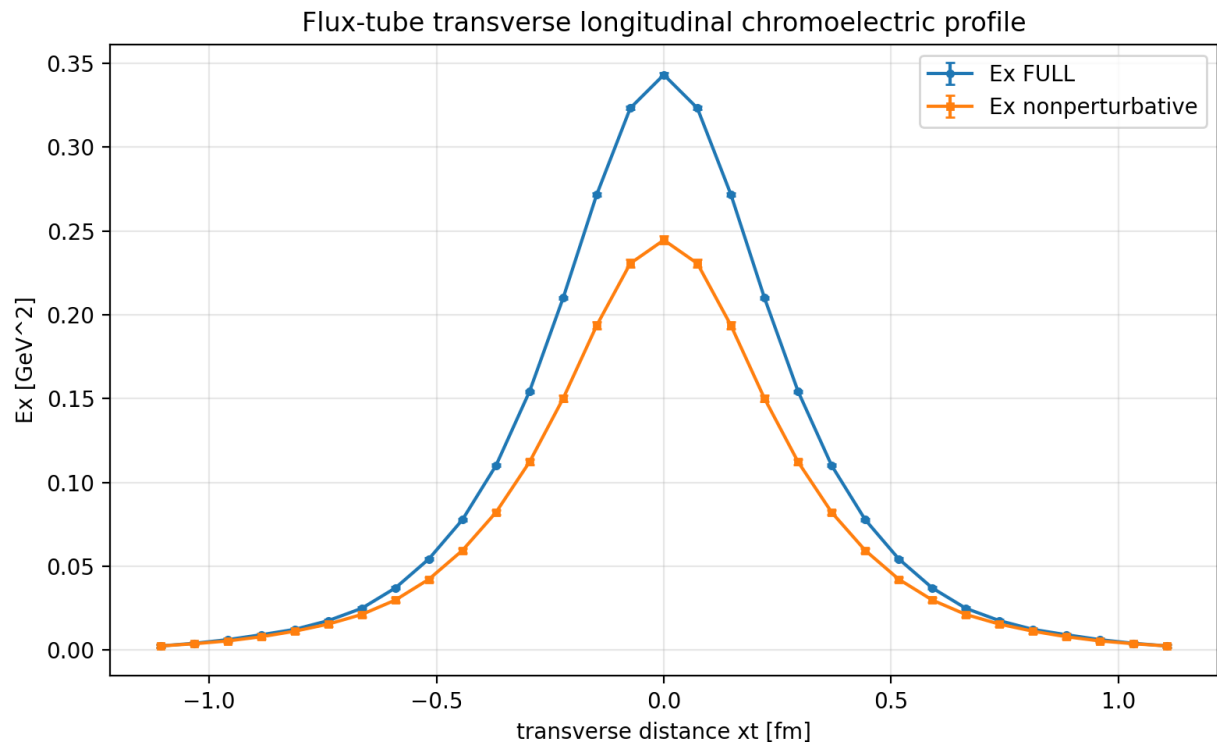
The static-source energy levels supply the first route-extension diagnostic.

Figure 2. Static-source level gaps



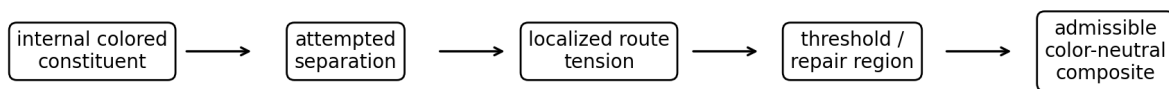
The gap plot is the first threshold diagnostic: competing levels approach and reorganize near the repair window.

Figure 3. Flux-tube transverse longitudinal chromoelectric profile



The profile supplies a direct geometry of route localization.

Figure 4. UNNS confinement sequence



UNNS working map: route extension -> boundary pressure -> repair into closed admissible state

Working map from internal constituent to admissible closed composite.

3. UNNS working map

QCD diagnostic object	UNNS interpretation
quark separation r	route-extension coordinate
$V_0(r)$, $V_1(r)$, $V_2(r)$	boundary-pressure / route-tension spectrum
string-breaking threshold	repair-threshold marker
two-meson threshold / screened channel	repaired admissible composite channel
flux-tube transverse profile $E(x_t)$	localized route geometry
absence of free color in ordinary external spectrum	non-externalizable internal coordinate

4. Formal interpretation

The first diagnostic supports a bounded claim: color confinement is a strong benchmark for the UNNS distinction between internal reality and external admissibility.

A colored constituent may be valid as an internal coordinate of the physical structure, but the isolated colored object is not the admissible external output. Attempted separation is carried by a localized, tensioned route. At threshold, the system is not interpreted as revealing free color; instead, the relevant channel is repaired into color-neutral composite structure.

5. Supported now

Static-source separation can be used as a first route-extension coordinate; the reconstructed static spectrum can be used as a provisional route-tension / boundary-pressure proxy; the reported light and strange string-breaking thresholds identify a repair-threshold window; and the Baker pointwise flux-tube profile gives direct localized-route geometry.

6. Not supported yet

This report does not provide a derivation of QCD confinement from UNNS. It does not contain raw measured V_0 , V_1 , V_2 lattice GEVP point tables, full uncertainty propagation for the reconstructed static spectrum, or a general law across QCD, H-mode, and other confinement systems.

7. Next report

The next formal report should be 02_repair_threshold_analysis.md, focused on r near $r_c \approx 1.224$ fm and r near $r_{cs} \approx 1.293$ fm, gap behavior, two-meson thresholds, and the transition from stretched route to repaired composite route.

8. Report decision

This diagnostic is suitable to enter the UNNS regime-synthesis track as: Color Confinement - Admissibility-by-Closure Benchmark I. Its function is not to prove QCD confinement, but to provide a grounded physical case where internal constituent, non-externalizable route, localized boundary pressure, and admissible closed composite can be studied quantitatively.